Probability, Statistics & Information Theory

Foundations for Machine Learning

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Basic Probability Terminologies

Key Definitions

- **Sample Space (S):** Set of all possible outcomes of an experiment.
- **Event (E):** Any subset of the sample space, $E \subseteq S$.
- **Random Experiment:** A process that produces uncertain outcomes.
- **Trial:** One single performance of the random experiment.
- **Mutually Exclusive Events:** $P(A \cap B) = 0$ — cannot occur simultaneously.
- **Exhaustive Events:** Union of all events covers the entire sample space.
- **Independent Events:** $P(A \cap B) = P(A) \cdot P(B)$

Basics of Probability

Classical Definition

For an event E in sample space S:

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes in } S}$$

- Range: $0 \leq P(E) \leq 1$
- $P(\emptyset) = 0, \quad P(S) = 1$
- Larger values indicate higher likelihood of occurrence.

Example

Rolling a fair die: $P(\text{getting a } 4) = \frac{1}{6} \approx 0.167$

Conditional Probability

Definition

Probability of event A given B has occurred:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

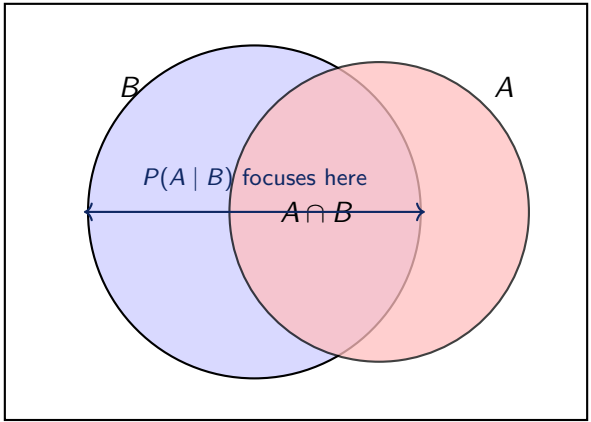
Numerical Example

- $A =$ "Student passed Maths", $B =$ "Student passed Physics"
- $P(A \cap B) = 0.30, \quad P(B) = 0.50$

$$P(A \mid B) = \frac{0.30}{0.50} = \mathbf{0.60}$$

- 60% of Physics passers also passed Maths.

Conditional Probability Visualization



Sample Space S

Rules of Probability

Complement Rule

$$P(A^c) = 1 - P(A)$$

Multiplication Rule

$$P(A \cap B) = P(A)\,P(B \mid A)$$

Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Independence

If $A \perp B$:

$$P(A \cap B) = P(A)\,P(B)$$

Random Variables

Definition

A random variable X maps outcomes of a sample space to real numbers: $X : S \to \mathbb{R}$

Discrete RV

- Takes countable values
- Described by PMF: $P(X = x)$

$$X = \text{die face}, \; P(X = x) = \frac{1}{6}$$

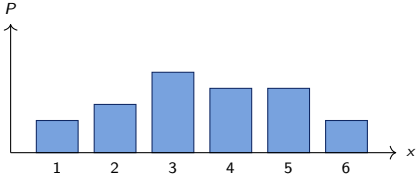
Continuous RV

- Takes values over an interval
- Described by PDF: $f(x)$

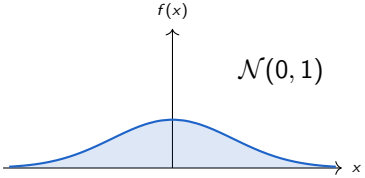
$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

PMF and PDF Illustration

Discrete PMF (Dice)



Continuous PDF $\mathcal{N}(0,1)$



Probability Distribution Table (Discrete)

Definition

A **Probability Distribution Table** lists every possible value of a discrete RV X alongside its probability $P(X = x)$, where $\sum_x P(X = x) = 1$.

Example — Number of Heads in 3 Coin Tosses

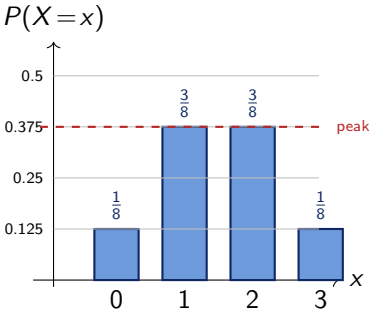
x (Heads)	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$
Decimal	0.125	0.375	0.375	0.125
$F(x) = P(X \leq x)$	0.125	0.500	0.875	1.000

Check: $\sum P(X = x) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1$ ✓

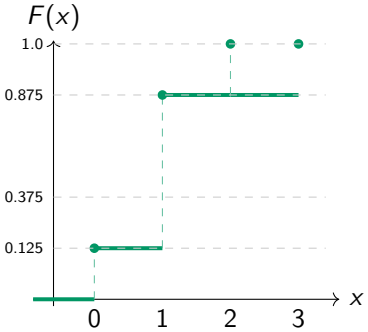
$F(x) =$ Cumulative Distribution Function (CDF)

PMF Visualization — 3 Coin Tosses

PMF: $P(X = x)$



CDF: $F(x) = P(X \leq x)$

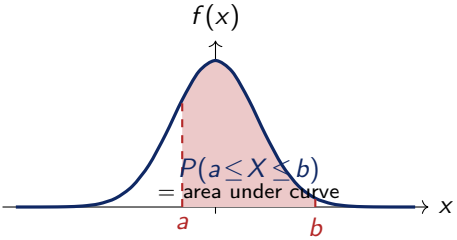


Probability Distribution Function (PDF) — Continuous

Definition

For a **continuous** RV X, the PDF $f(x)$ satisfies:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx, \qquad f(x) \geq 0, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1$$



Key Differences from Discrete:

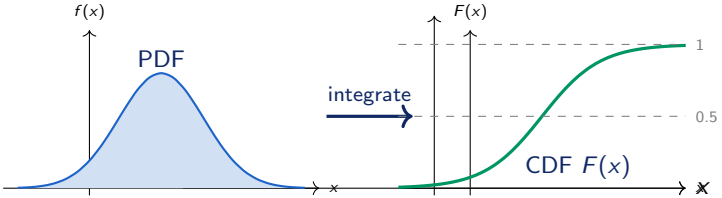
- $P(X = x) = 0$ for any single point
- Probability is the **area** under $f(x)$
- $f(x)$ can exceed 1 (it's a density, not a probability)
- CDF: $F(x) = \int_{-\infty}^{x} f(t)\,dt$
- $f(x) = \dfrac{d}{dx}F(x)$

CDF of Continuous Distribution

Cumulative Distribution Function

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\, dt$$

$F(x)$ is a non-decreasing function from 0 to 1.



Discrete vs Continuous: Summary Comparison

Property	Discrete RV	Continuous RV
Distribution	PMF $P(X = x)$	PDF $f(x)$
Point prob.	$P(X = x) > 0$	$P(X = x) = 0$
Range prob.	$\sum_{x=a}^{b} P(X = x)$	$\int_a^b f(x)\,dx$
CDF	$F(x) = \sum_{t \leq x} P(t)$	$F(x) = \int_{-\infty}^{x} f(t)\,dt$
Sum / Integral	$\sum_x P(X = x) = 1$	$\int_{-\infty}^{\infty} f(x)\,dx = 1$
Example	Dice rolls, coin tosses	Height, temperature

Measures of Central Tendency

Mean (μ)

Population: $\mu = \frac{1}{N} \sum_{i=1}^{N} x_i$

Weighted: $\mu = \sum x_i \cdot P(x_i)$

Median

Middle value when data is sorted.

For n odd: $x_{(n+1)/2}$

For n even: $\frac{x_{n/2} + x_{n/2+1}}{2}$

Mode

Most frequently occurring value.

A distribution can be:

- *Unimodal* — one peak
- *Bimodal* — two peaks
- *Multimodal* — many peaks

Example

Data: 2, 4, 4, 6, 8

Mean = 4.8, Median = 4, Mode = 4

Measures of Dispersion

Variance (σ^2) and Standard Deviation (σ)

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2, \qquad \sigma = \sqrt{\sigma^2}$$

Sample variance uses $N - 1$ (Bessel's correction): $s^2 = \frac{1}{N-1}\sum(x_i - \bar{x})^2$

Interpretation:

- Small $\sigma \Rightarrow$ data clustered near mean
- Large $\sigma \Rightarrow$ data spread out
- σ has same units as data

Example

Data: 2, 4, 4, 6, 8

$\mu = 4.8$

$\sigma^2 = 4.56$

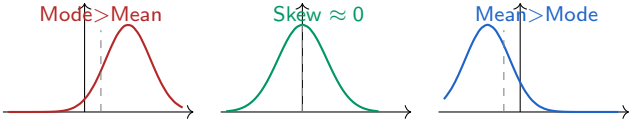
$\sigma \approx 2.135$

Skewness

Definition

Skewness measures the **asymmetry** of a distribution:

$$\text{Skewness} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i - \mu}{\sigma}\right)^3$$



Kurtosis

Definition

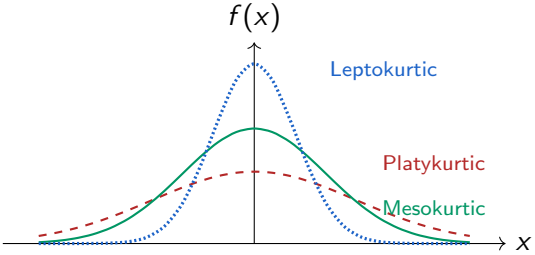
Kurtosis measures **tail heaviness** (peakedness) of a distribution:

$$\text{Kurtosis} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i - \mu}{\sigma}\right)^4$$

Excess Kurtosis = Kurtosis − 3 (Normal distribution has excess kurtosis = 0)

Type	Excess Kurtosis	Tails
Leptokurtic	> 0	Heavier than normal (fat tails)
Mesokurtic	$= 0$	Same as normal
Platykurtic	< 0	Lighter than normal (thin tails)

Kurtosis Visualization



All three curves have the same mean and variance.

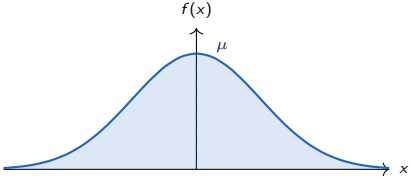
Normal Distribution

Probability Density Function

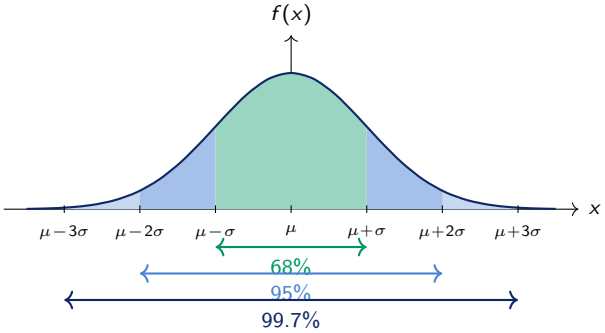
$$f(x \mid \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Key Properties:

- Bell-shaped and symmetric
- Mean = Median = Mode = μ
- Skewness = 0, Kurtosis = 3
- 68–95–99.7 Rule
- Fully defined by μ and σ



The 68–95–99.7 Rule (Empirical Rule)



Standard Normal Distribution & Z-Score

Standardization

Any normal variable $X \sim \mathcal{N}(\mu, \sigma^2)$ can be standardized:

$$Z = \frac{X - \mu}{\sigma}, \qquad Z \sim \mathcal{N}(0, 1)$$

Numerical Example

Given: Exam scores $\sim \mathcal{N}(70,\ 10^2)$.
Find the probability that a student scores above 85.

$$Z = \frac{85 - 70}{10} = 1.5$$

$$P(X > 85) = P(Z > 1.5) = 1 - \Phi(1.5) = 1 - 0.9332 = \mathbf{0.0668}$$

About **6.68%** of students score above 85.

Bayes' Theorem – Limited Form

Bayes' Theorem

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

- $P(A \mid B)$ — **Posterior:** probability of A after observing B
- $P(B \mid A)$ — **Likelihood:** how likely B is if A is true
- $P(A)$ — **Prior:** initial belief about A
- $P(B)$ — **Evidence:** normalizing constant

Note

This two-event form applies without partitioning the sample space.

Bayes' Theorem (General Form)

With Partition $\{B_1, B_2, \ldots, B_n\}$ of S

$$P(B_i \mid A) = \frac{P(A \mid B_i)\, P(B_i)}{\sum_{j=1}^{n} P(A \mid B_j)\, P(B_j)}$$

Applications:

- Medical diagnosis (disease testing)
- Spam email filtering
- Machine learning classifiers
- Fault detection in systems

Bayes' Theorem: Numerical Example

Disease Testing Problem:

- $P(D) = 0.01$ (1% prevalence), $P(\bar{D}) = 0.99$
- $P(T^+ \mid D) = 0.99, \quad P(T^+ \mid \bar{D}) = 0.01$

Solution — What is $P(D \mid T^+)$?

$$P(T^+) = (0.99)(0.01) + (0.01)(0.99) = 0.0198$$

$$P(D \mid T^+) = \frac{P(T^+|D)\,P(D)}{P(T^+)} = \frac{0.99 \times 0.01}{0.0198} = \mathbf{0.50}$$

Insight

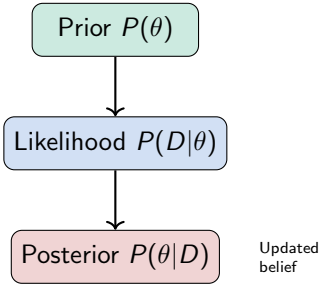
Even with a 99% accurate test, there is only a 50% chance of disease given a positive result when prevalence is low — the base rate matters!

Bayesian Inference View

- **Prior** $P(\theta)$: belief before data
- **Likelihood** $P(D|\theta)$: data fit under θ
- **Posterior** $P(\theta|D)$: updated belief

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

- Denominator $P(D)$ is often intractable $\Rightarrow$ MCMC, Variational Inference



Entropy – Theory

Shannon Entropy

For a discrete random variable X with outcomes $\{x_1, \ldots, x_n\}$ and probabilities $p_i = P(X = x_i)$:

$$H(X) = -\sum_{i=1}^{n} p_i \log_2 p_i \quad \text{(bits)}$$

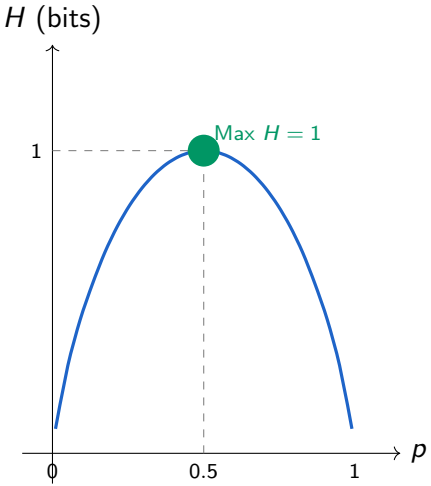
Intuition:

- Entropy = average **surprise** or **uncertainty**
- High entropy $\Rightarrow$ more uncertainty
- Low entropy $\Rightarrow$ more predictable
- $H = 0$ when outcome is certain
- H maximized by uniform distribution

Special Cases

- Fair coin: $H = 1$ bit
- Biased coin ($p = 1$): $H = 0$
- Fair die: $H = \log_2 6 \approx 2.58$ bits

Entropy Visualization



$$H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$$

Binary entropy $H(p)$ — peaks at $p = 0.5$ (maximum uncertainty)

Entropy: Numerical Example

Example — Weather Forecasting

Probabilities: Sunny $= 0.5$, Cloudy $= 0.25$, Rainy $= 0.25$

$$H = -\sum p_i \log_2 p_i$$

$$\begin{aligned} H &= -\left[0.5\log_2(0.5) + 0.25\log_2(0.25) + 0.25\log_2(0.25)\right] \\ &= -\left[0.5(-1) + 0.25(-2) + 0.25(-2)\right] \\ &= -\left[-0.5 - 0.5 - 0.5\right] \\ &= \mathbf{1.5} \text{ bits} \end{aligned}$$

Interpretation

The system requires 1.5 bits on average to encode the weather outcome. A uniform distribution (all equal prob.) would give $H = \log_2 3 \approx 1.585$ bits.

Conditional Entropy & Information Gain

Conditional Entropy

$$H(Y \mid X) = \sum_x P(X = x) \cdot H(Y \mid X = x)$$

Uncertainty remaining in Y after observing X.

Information Gain (IG)

$$IG(Y, X) = H(Y) - H(Y \mid X)$$

Reduction in entropy of Y after knowing X.

- $IG \geq 0$ always (knowing something can't increase uncertainty)
- $IG = 0$: X is irrelevant to Y
- $IG = H(Y)$: X completely determines Y
- Used in **Decision Tree** algorithms (ID3, C4.5)

Information Gain: Numerical Example

Training data — 14 days: 9 Play, 5 No-Play

$$H(\text{Play}) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} \approx \mathbf{0.940}$$

Split by Outlook:

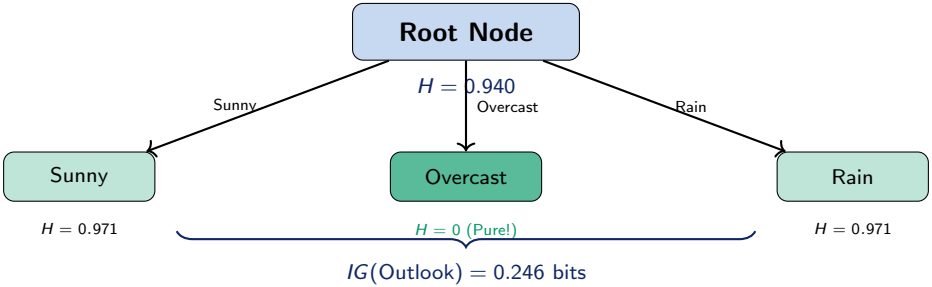
Outlook	Count	Play	H
Sunny	5	2 Yes, 3 No	0.971
Overcast	4	4 Yes, 0 No	0.000
Rain	5	3 Yes, 2 No	0.971

$$H(\text{Play} \mid \text{Outlook}) = \frac{5}{14}(0.971) + \frac{4}{14}(0.000) + \frac{5}{14}(0.971) \approx 0.694$$

$$IG = 0.940 - 0.694 = \mathbf{0.246} \text{ bits}$$

Splitting on *Outlook* reduces uncertainty by 0.246 bits — choose the attribute with highest IG!

Information Gain Visualization



Summary

Key Takeaways

- **Probability** provides the mathematical framework for uncertainty.
- **Statistical properties** (mean, median, mode, variance, skewness, kurtosis) characterize distributions fully.
- **Normal distribution** is the cornerstone of statistical inference — symmetric, fully described by μ and σ.
- **Bayes' Theorem** enables principled updating of beliefs with evidence.
- **Entropy** quantifies uncertainty; **Information Gain** measures how much a feature reduces it.
- IG is the foundation for **decision tree** learning algorithms.

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